Lecture 15: Integer Representations

Section 4.2

Section Summary

- Integer Representations
  - Base *b* Expansions
  - Binary Expansions
  - Octal Expansions
  - Hexadecimal Expansions
- Base Conversion Algorithm
- Algorithms for Integer Operations

Representations of Integers

- In the modern world, we use *decimal,* or *base* 10, *notation* to represent integers. For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.
- We can represent numbers using any base b, where b is a positive integer greater than 1.
- The bases $b = 2$ (*binary*), $b = 8$ (*octal*) , and $b = 16$ (*hexadecimal*) are important for computing and communications
- The ancient Mayans used base 20 and the ancient Babylonians used base 60.

Base *b* Representations

- We can use positive integer *b* greater than 1 as a base, because of this theorem:

  Theorem 1 : Let *b* be a positive integer greater than 1. Then if *n* is a positive integer, it can be expressed uniquely in the form:

  $$n = a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0$$

  where *k* is a nonnegative integer;

  $a_0, a_1, \ldots a_k$ are nonnegative integers less than *b*, and $a_k \neq 0$.

  The a_j terms with $j = 0,\ldots,k$ are called the **base- *b* digits** of the representation.

- The representation of *n* given in Theorem 1 is called the ***base b expansion of n*** and is denoted by $(a_k a_{k-1} \ldots a_1 a_0)_b$.

Base b Representations

From the division algorithm, we can write

$n = b \cdot q_0 + r_0$; where $0 \le r_0 < b$ and $q_0 < n$.

Now, q_0 can also be written as: $q_0 = b \cdot q_1 + r_1$, where $q_1 < q_0$.

and consequently, $q_1 = b \cdot q_2 + r_2$, where $q_2 < q_1$, and so on.

Eventually, after k steps, $\boldsymbol{q_k}$ **will be zero** . So, $q_{k-1} = b \cdot 0 + r_k = r_k$.

Thus we have,

$$n = b \cdot q_0 + r_0 = b \cdot (b \cdot q_1 + r_1) + r_0 = b^2 \cdot q_1 + b \cdot r_1 + r_0$$

$$= b^2 \cdot (b \cdot q_2 + r_2) + b \cdot r_1 + r_0 = b^3 \cdot q_2 + b^2 \cdot r_2 + b \cdot r_1 + r_0$$

$$= b^{k+1} \cdot \boldsymbol{q_k} + b^k \cdot r_k + b^{k-1} \cdot r_{k-1} + \ldots\ldots + b^3 \cdot r_3 + b^2 \cdot r_2 + b \cdot r_1 + r_0$$

$$= b^k \cdot r_k + b^{k-1} \cdot r_{k-1} + \ldots\ldots + b^3 \cdot r_3 + b^2 \cdot r_2 + b \cdot r_1 + r_0 \; [q_k = 0]$$

Binary Expansions

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

Example : What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?

Solution : $(1\ 0101\ 1111)_2$

$= 1\cdot 2^8 + 0\cdot 2^7 + 1\cdot 2^6 + 0\cdot 2^5 + 1\cdot 2^4 + 1\cdot 2^3 + 1\cdot 2^2 + 1\cdot 2^1 + 1\cdot 2^0 = 351.$

Example : What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution : $(11011)_2 = 1\cdot 2^4 + 1\cdot 2^3 + 0\cdot 2^2 + 1\cdot 2^1 + 1\cdot 2^0 = 27.$

Octal Expansions

The octal expansion (base 8) uses the digits {0,1,2,3,4,5,6,7}.

Example : What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution : $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Example : What is the decimal expansion of the number with octal expansion $(111)_8$?

Solution : $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

Hexadecimal Expansions

The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}. The letters A through F represent the decimal numbers 10 through 15.

Example : What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

Solution :

$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$

Example : What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

Solution : $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$

Base Conversion

Example : Find the octal expansion of $(12345)_{10}$

Solution : Successively dividing by 8 gives:

$$12345 = 8 \cdot 1543 + 1$$
$$1543 = 8 \cdot 192 + 7$$
$$192 = 8 \cdot 24 + 0$$
$$24 = 8 \cdot 3 + 0$$
$$3 = 8 \cdot 0 + 3$$

The remainders are the digits from right to left, yielding $(30071)_8$.

Comparison of Hexadecimal, Octal, and Binary Representations

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Initial 0s are not shown

- Each octal digit corresponds to a block of 3 binary digits.
- Each hexadecimal digit corresponds to a block of 4 binary digits.
  So, conversion between binary, octal, and hexadecimal is easy.

Conversion Between Binary, Octal, and Hexadecimal Expansions

Example : Find the octal and hexadecimal expansions of $(11\ 1110\ 1011\ 1100)_2$.

Solution :

- To convert to octal, we group the digits into blocks of three, adding initial 0s as needed:

  $(011\ \ 111\ \ 010\ \ 111\ \ 100)_2$

  The blocks from left to right correspond to the digits 3, 7, 2, 7 and 4. Hence, the solution is $(37274)_8$.

- To convert to hexadecimal, we group the digits into blocks of four, adding initial 0s as needed:

  $(0011\ \ 1110\ \ 1011\ \ 1100)_2$

  The blocks from left to right correspond to the digits 3, E, B and C. Hence, the solution is $(3EBC)_{16}$.

Binary (Fast) Modular Exponentiation

In cryptography, it is important to be able to find $b^n \bmod m$ efficiently, where b, n, and m are large integers.

- We can compute the exponential using the fact:

  $$b^{s+t} \bmod m = (b^s \bmod m)(b^t \bmod m) \bmod m$$

- Also, binary representation of n yields:

  $$n = a_{k-1}2^{k-1} + \ldots + a_1 2 + a_0 = (a_{k-1} \ldots a_1\, a_0)_2$$

  - Note that Here $a_k, a_{k-1}, \ldots, a_1, a_0 \in \{0, 1\}$.

- Then,

  $$b^n = b^{a_{k-1}\cdot 2^{k-1}+\cdots+a_1\cdot 2+a_0} = b^{a_{k-1}\cdot 2^{k-1}} \ldots b^{a_1\cdot 2} \cdot b^{a_0}$$

- To calculate $b^n \bmod m$, successively find:

  $b \bmod m$, $b^2 \bmod m$, $b^4 \bmod m, \ldots, b^{2\wedge(k-1)} \bmod m$

  Then, multiply together the terms $b^{2\wedge j} \bmod m$ where $a_j = 1$.

Example : Compute 3^{11} using this method.

Solution : Note that $11 = (1011)_2$ so that $3^{11} = 3^8\, 3^2\, 3^1 = ((3^2)^2)^2\, 3^2\, 3^1$ $= (9^2)^2 \cdot 9 \cdot 3 = (81)^2 \cdot 9 \cdot 3 = 6561 \cdot 9 \cdot 3 = 117{,}147$.

Fast Modular Exponentiation Example

Example : Find 3^{144} **mod** 100.

Solution : Here, $144 = (10010000)_2$ so $3^{144} = 3^{128}\ 3^{16}$. Now,

- 3 **mod** $100 = 3$
- 3^2 **mod** $100 = (3 * 3)$ **mod** $100 = 9$
- 3^4 **mod** $100 = (9 * 9)$ **mod** $100 = 81$
- 3^8 **mod** $100 = (81 * 81)$ **mod** $100 = 6561$ **mod** $100 = 61$
- 3^{16} **mod** $100 = (61 * 61)$ **mod** $100 = 3721$ **mod** $100 = 21$
- 3^{32} **mod** $100 = (21 * 21)$ **mod** $100 = 441$ **mod** $100 = 41$
- 3^{64} **mod** $100 = (41 * 41)$ **mod** $100 = 1681$ **mod** $100 = 81$
- 3^{128} **mod** $100 = (81 * 81)$ **mod** $100 = 6561$ **mod** $100 = 61$

So, 3^{144} **mod** $100 = (3^{128}$ **mod** $100)(3^{16}$ **mod** $100)$ **mod** 100

$= (61*21)$ **mod** $100 = 1281$ **mod** $100 = 81$.

Therefore, 3^{144} **mod** $10 = 81$ (Answer)